IC4SD-ForensicMacroWood-CITES: A Macroscopic Transverse Image Benchmark with Species-Level Anatomical Descriptors for CITES-Regulated Hardwood Identification

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ABSTRACT

Rapid forensic discrimination of protected timbers at international checkpoints requires standardized macroscopic reference datasets. We present IC4SD-ForensicMacroWood-CITES, an authenticated benchmark comprising 6,414 standardized transverse end-grain image captures (224 × 224 px) across 19 commercially significant Fabaceae hardwoods (11 CITES Appendix II-regulated taxa and 8 non-regulated commercial controls), anchored to 147 verified physical wood block accessions. The dataset integrates an expert-curated species-level reference matrix of 14 macroscopic anatomical descriptors standardized under IAWA criteria, providing structured morphological domain knowledge for taxonomic attribute grounding and text-guided classification. To address reference specimen scarcity, we establish a dual-partition architecture: an operational canonical benchmark maintaining 100% class coverage alongside an audited specimen-disjoint benchmark (0.0% specimen leakage across multi-specimen taxa). Technical validation using ResNet-50 confirms reproducible baseline performance, achieving 88.64% overall accuracy on the operational canonical split and 80.12% on the strict specimen-disjoint split, highlighting an empirical generalization gap across unseen logs. Frontline customs triage at the genus level reaches 94.37% accuracy. The benchmark is openly accessible under CC BY 4.0 at DOI: 10.5281/zenodo.14892180.

Specifications Table

Subject	Computer Science; Forestry and Wood Science
Specific subject area	Forensic Wood Anatomy, CITES Trade Compliance, Computer Vision, Representation Learning, Statistical Validation
Type of data	Image (standardized 224×224 px RGB JPEG, $Q = 95$, in `images.zip`), Table (master CSV metadata, split manifests, structured 14-axis anatomical descriptors in `metadata/anatomical_features.csv`), Mapping (JSON label map)
Data format	Raw, filtered, standardized RGB images (JPEG), structured tabular metadata (CSV), label mapping dictionaries (JSON)
Data collection	Transverse end-grain surfaces surfaced orthogonally on a sliding table saw, progressively polished across four grit gradations (P120–P600), cleaned with dry air jets to clear sawdust residues, and captured under daylight-balanced (5600 K) diffuse circular LED illumination at fixed optical magnification to extract standardized 224×224 px non-overlapping macroscopic image tiles
Data source location	**Institution:** Intelligent Computing for Sustainable Development Laboratory (IC4SD), Posts and Telecommunications Institute of Technology (PTIT) **City/Country:** Hanoi, Vietnam
Data accessibility	**Repository:** Zenodo Scientific Archive **Data identification number:** 10.5281/zenodo.14892180 **Direct URL to data:** `https://doi.org/10.5281/zenodo.14892180` **Instructions for accessing these data:** All files, split manifests, anatomical matrices, and the executable demonstration script (`code/quickstart_demo.py`) are openly downloadable without restrictions under CC BY 4.0 license
Related research article	None. (A companion research paper exploring advanced vision-language grounding and domain generalization models is in preparation).

1. Value of the Data

- **Dedicated Forensic Reference for CITES Hardwood Compliance**: This dataset delivers an authenticated macroscopic benchmark specifically engineered for fine-grained forensic discrimination of 19 commercially significant tropical Fabaceae hardwoods. With 11 taxa strictly regulated under CITES Appendix II (incorporating African populations of *Afzelia* and *Pterocarpus* under CoP19 Annotation #17 alongside the universal *Dalbergia* listing under CoP17 Annotation #15), it directly equips customs inspection checkpoints, CITES Scientific and Management Authorities, and wildlife enforcement agencies (INTERPOL, UNODC) with a diagnostic tool against illegal timber trafficking.
- **Rapid Non-Destructive Frontline Customs Triage**: In contrast to classical forensic microtomy which requires days of laboratory softening, sectioning, and staining, or costly molecular barcoding and mass spectrometry (DART-TOFMS), these standardized transverse macroscopic captures (224×224 px) establish a rapid screening baseline to assist in verifying containerized timber consignments under CITES and timber trade compliance scrutiny.

- **Structured Anatomical Ground Truth for Attribute-Guided AI**: The benchmark bridges computer vision and classical xylotomy through an expert-curated matrix of 14 macroscopic anatomical descriptors standardized under IAWA criteria. This structural metadata serves as an objective species-level attribute reference for taxonomic attribute grounding and text-guided classification, enabling models to associate visual tissue features with authentic morphological criteria.
- **Dual-Split Architecture Addressing Specimen Scarcity Bottlenecks**: To address the acute specimen scarcity inherent to legally protected timber archives ($|\mathcal{G}_c| \leq 10$ physical blocks per taxon, with acute singleton accessions), the repository establishes a dual-partition architecture: an operational canonical benchmark providing 100% Class Coverage Rate across all subsets for frontline inspection, and an audited specimen-disjoint split (0.0% specimen leakage across 17 multi-specimen taxa) to quantify model invariance against genuine out-of-distribution biological variance.
- **Turnkey Machine Learning Reproducibility**: The benchmark provides scale-preserving cropped tiles alongside systematic multi-seed classification baselines, establishing empirical performance standards and modular code pipelines to facilitate adoption across forestry informatics and computer vision communities.

2. Data Description

2.1. Taxonomic Composition and Inventory

The IC4SD-ForensicMacroWood-CITES repository comprises **6,414 standardized macroscopic cross-sectional captures** spanning **19 commercially significant tropical hardwood species** across **6 botanical genera** within the legume family **Fabaceae** (Leguminosae). All images were curated from institutional timber reference archives at the Intelligent Computing for Sustainable Development Laboratory (IC4SD), PTIT, Hanoi, Vietnam. Each capture carries a verified species-level label linked to traceable physical wood specimens ($|\mathcal{G}| = 147$ verified wood block accessions) and confirmed through an independent dual-expert verification protocol ($\kappa = 0.985$).

Table 1 details the taxonomic breakdown of the 19 species, providing scientific nomenclature, native Vietnamese vernacular terms, international trade designations, CITES conservation listings, physical specimen cohort sizes ($|\mathcal{G}_c|$, totaling 147 verified wood block accessions), and total image volumes.

2.2. Comparative Dataset Positioning

To contextualize the forensic utility of IC4SD-ForensicMacroWood-CITES, Table 2 systematically compares its core attributes against prominent public macroscopic wood datasets and recent literature benchmarks [1–6]. Foundational initiatives like the XyloTron project [1, 2] and regional surveys [3, 4] have significantly advanced computer vision in forestry. However, these existing public collections typically focus on non-regulated commercial timbers with minimal CITES coverage ($< 15\%$). They also frequently utilize lower optical magnifications or non-standardized mobile lenses.

Recent institutional benchmarks from our laboratory, such as IC4SD-Wood-Eucalyptus [6] (our previous work), established systematic image curation but focused exclusively on plantation hardwoods (0% CITES) for pulpwood management. Meanwhile, specialized studies targeting high-value CITES taxa [5] remain restricted to private, unshared archives. This highlights a critical gap in open-access forensic resources for endangered tropical hardwoods.

IC4SD-ForensicMacroWood-CITES addresses this gap with three distinguishing contributions. First, it targets **19 commercially significant tropical hardwoods across 6 Fabaceae genera**, with a critical concentration of trade-restricted timbers (**57.9% CITES Appendix II**, 11 regulated species) evaluated alongside **8 non-regulated commercial controls and lookalikes** (comprising 4 high-value regional commercial timbers and 4 international commercial species). Second, it provides an expert-curated matrix of **14 diagnostic**

Table 1
Taxonomic inventory and physical specimen sampling of the 19 tropical hardwood species in the IC4SD-ForensicMacroWood-CITES benchmark, detailing botanical genus, scientific binomial nomenclature, native Vietnamese vernacular names, international trade names, CITES conservation status, verified physical specimen cohort sizes ($|\mathcal{G}_c|$, totaling 147 verified wood block accessions), and standardized image counts.

#	Botanical Genus	Scientific Binomial	Vietnamese Vernacular	Trade / Common Name	CITES Status	Physical Specimens ($\|\mathcal{G}_c\|$)	Total Images
1	*Afzelia*	*Afzelia africana*	Gõ Douse (Gõ Doussié)	African Doussié	CITES App. II[a]	4	241
2	*Afzelia*	*Afzelia bella*	Papao-Nua / Gỗ Gõ	Bella Doussié	CITES App. II[a]	10	400
3	*Afzelia*	*Afzelia pachyloba*	Gõ Pachy	White Doussié	CITES App. II[a]	5	116
4	*Afzelia*	*Afzelia quanzensis*	Gõ Quanzensis	Pod Mahogany	CITES App. II[a]	8	369
5	*Dalbergia*	*Dalbergia cochinchinensis*	Trắc (Rosewood)	Siam Rosewood	CITES App. II[b] (Native / High-value regional)	1	354
6	*Dalbergia*	*Dalbergia melanoxylon*	Trắc châu Phi	African Blackwood	CITES App. II[b]	10	291
7	*Dalbergia*	*Dalbergia oliveri*	Cẩm lai (Burmese Rosewood)	Burmese Rosewood	CITES App. II[b]	10	316
8	*Dalbergia*	*Dalbergia rimosa*	Trắc dây	Rimose Rosewood	CITES App. II[b]	10	300
9	*Dalbergia*	*Dalbergia tonkinensis*	Sưa	Vietnamese Rosewood	CITES App. II[b] (Native / High-value regional)	10	325
10	*Guibourtia*	*Guibourtia arnoldiana*	Gỗ Muntenye	Mutenye / Benge	Non-CITES[c] (International commercial)	10	323
11	*Guibourtia*	*Guibourtia coleosperma*	Mussivi / Hương đá	Rhodesian Copalwood	Non-CITES[c] (International commercial)	1	360
12	*Guibourtia*	*Guibourtia ehie*	Hyedua	Ovangkol / Shedua	Non-CITES[c] (International commercial)	10	400
13	*Peltogyne*	*Peltogyne pubescens*	Hương tím nam mỹ	Purpleheart	Non-CITES (International commercial)	10	371
14	*Pterocarpus*	*Pterocarpus erinaceus*	Hương vân tây phi	African Barwood / Kosso	CITES App. II[d]	10	336
15	*Pterocarpus*	*Pterocarpus indicus*	Hương mắt chim	Narra / Amboyna	Non-CITES (Native / High-value regional)	10	312
16	*Pterocarpus*	*Pterocarpus macrocarpus*	Hương quả to	Burma Padauk	Non-CITES (Native / High-value regional)	8	432
17	*Pterocarpus*	*Pterocarpus soyauxii*	Padouk / Hương padouk	African Padauk	CITES App. II[d]	6	486
18	*Sindora*	*Sindora cochinchinensis*	Gụ	Sindora / Sepetir	Non-CITES (Native / High-value regional)	4	354
19	*Sindora*	*Sindora tonkinensis*	Gụ lau	Tonkin Sepetir	Non-CITES (Native / High-value regional)	10	328
Total Benchmark	**19 Botanical Species (6 Genera, 11 CITES App. II, 8 Non-CITES)**			**Total**	**147**	**6,414**

[a]African populations of the genus *Afzelia* (*A. africana*, *A. bella*, *A. pachyloba*, *A. quanzensis*) were included in CITES Appendix II with annotation #17 (logs, sawn wood, veneer sheets, plywood and transformed wood) at CoP19 (Panama City, 2022) with entry into force on 23 February 2023. *A. pachyloba* constitutes a statistical minority class ($N = 116$ captures, 5 physical blocks) due to acute reference specimen scarcity. [b]All species of the genus *Dalbergia* (with the exception of *D. nigra* listed in Appendix I) are regulated under CITES Appendix II with annotation #15 since CoP17 (2017). [c]All three examined species of the genus *Guibourtia* (*G. arnoldiana*, *G. coleosperma*, *G. ehie*) are non-CITES commercial timbers, distinct from the three CITES-listed bubinga species (*G. demeusei*, *G. pellegriniana*, *G. tessmannii*); they serve as essential negative controls for forensic compliance. [d]African populations of the genus *Pterocarpus* (including *P. soyauxii* and *P. erinaceus*) are regulated under CITES Appendix II with annotation #17 (logs, sawn wood, veneer sheets, plywood and transformed wood), which entered into force on 23 February 2023 following CoP19 decisions; *P. erinaceus* had been included in Appendix II since 2016.

macroscopic anatomical descriptors structured under IAWA xylotomical standards, enabling structured morphological attribute grounding absent in prior single-modality collections. Third, it delivers standardized non-overlapping macro captures (224×224 px), verified physical block traceability, dual partition manifests, and audited bitwise integrity boundaries.

Table 2
Systematic comparison of IC4SD-ForensicMacroWood-CITES against prominent public and literature macroscopic timber datasets, contrasting taxonomic focus, optical magnification, forensic regulation, partition governance, species-level anatomical descriptors, and machine learning accessibility.

Dataset / Study	Taxonomic Focus	No. Spp.	Total Images	Optical Mag. / Resolution	CITES App. II Ratio (%)	Partition Protocol	Contamination Audit (SHA-256 / Hash)	Species-Level IAWA Descriptors	Access Model
XyloTron Datasets [1, 2]	Neotropical/US hardwoods	10–40	2,300–6,000	10× (24 μm/px)	Low (< 15%)	Specimen-level split	Not reported	No	Open
Costa Rican Timbers [3]	Native Costa Rican trees	21	2,360	Variable (phone macro)	Minimal (< 5%)	Random image split	Not reported	No	Open
Asian Commercial [4]	Vietnamese/Asian timber	15	∼3,000	20×–40× USB	Moderate (20%)	Random k-fold	Not reported	No	Restricted
Pterocarpus Discrimination [5]	*Pterocarpus* genus only	6	∼1,800	20×–30× stereo	Subset (50%, 3 spp.)	Random split	Not reported	No	Private
IC4SD-Wood-Eucalyptus (our previous work) [6]	Plantation genus *Eucalyptus*	10	2,910	12.0 μm/px	None (0.0%, 0 spp.)	Specimen-aware (audited)	Bitwise SHA-256	No	Open (CC BY 4.0)
IC4SD-ForensicMacroWood-CITES (This Work)	**Tropical Fabaceae (6 genera)**	**19**	**6,414**	**Macro 224×224 px (fixed opt.)**	**High (57.9%, 11 spp.)**	**Dual (Canonical / Disjoint)**	**Bitwise SHA-256**	**Yes (14-axis CSV)**	**Open (CC BY 4.0)**

2.3. Repository Architecture and Metadata Schema

The dataset package is organized to facilitate straightforward integration into machine learning pipelines. Table 3 outlines the directory architecture, and Table 4 defines the structured metadata attributes in

`metadata.csv`. An executable walkthrough script (`code/quickstart_demo.py`) is also provided, allowing users to verify metadata schemas, inspect image quality scores, and replicate baseline classification evaluations within seconds.

Table 3
Systematic directory layout and functional contents of the IC4SD-ForensicMacroWood-CITES release package.

#	File / Folder Path	Content Description and Functionality
1	`images.zip`	Compressed ZIP archive containing 6,414 standardized 224×224 px macroscopic RGB images organized by taxon subfolders.
2	`metadata/`	Master metadata table (`metadata.csv`), structured anatomical descriptors (`anatomical_features.csv`), label mappings (`label_map.json`), and bitwise SHA-256 release manifest (`release_manifest.csv`).
3	`splits/`	Governed canonical partition manifest (`split_canonical.csv`; $N_{\text{train}} = 3,959$, $N_{\text{val}} = 1,265$, $N_{\text{test}} = 1,190$) alongside strict specimen-disjoint split manifest (`split_specimen_disjoint.csv`).
4	`code/`	Standalone demonstration script (`quickstart_demo.py`), interactive walkthrough notebook (`quickstart_demo.ipynb`), and dependencies (`requirements.txt`).
5	`classification_output/`	ResNet-50 supervised classification baseline checkpoints (`resnet50_focal_best.pth`), test predictions, and evaluation metrics under Focal Loss.
6	`README.md`, `LICENSE`	Comprehensive dataset documentation, quickstart instructions, and Creative Commons Attribution 4.0 International license terms.

2.4. Partition Allocations and Class Distributions

The 6,414 captures are partitioned following a governed specimen-aware allocation protocol across physical wood blocks (Training: 3,959 images, 61.72%; Validation: 1,265 images, 19.72%; Test: 1,190 images, 18.55%). All 19 species achieve a 100% Class Coverage Rate (CCR = 100.0%, 19/19 taxa) across all three partitions.

Authentic timber reference archives are inherently characterized by small specimen cohorts. Across all 19 taxa, our collection contains 147 verified physical wood block accessions ($|\mathcal{G}_c| \leq 10$ per species), with acute bottlenecks such as a single physical specimen accession for *Dalbergia cochinchinensis* ($N = 354$) and *Guibourtia coleosperma* ($N = 360$). Under these small-cohort constraints, dogmatic three-way whole-block isolation inevitably deprives minority CITES taxa of evaluation captures. For example, enforcing complete block isolation on a single-specimen class structurally precludes validation or test coverage.

To resolve this trade-off between operational coverage and out-of-distribution evaluation, our dataset architecture provides two distinct, explicitly documented partitions:

1. **Operational Canonical Benchmark** (`splits/split_canonical.csv`): Designed as a practical operational baseline for customs screening, prioritizing a 100% Class Coverage Rate (CCR = 100.0%, 19/19 taxa) across Train, Validation, and Test subsets through controlled specimen boundary sharing.

Table 4
Core metadata attributes and data schema documented within `metadata.csv`.

#	Field Name	Data Type	Field Description and Functionality
1	`image_id`	String	Unique standardized image identifier (e.g., `VNMW_000001`).
2	`image_path`	String	Relative file path to the macroscopic image within the repository archive.
3	`genus`	String	Botanical genus name (*Afzelia*, *Dalbergia*, *Guibourtia*, *Peltogyne*, etc.).
4	`species`	String	Botanical species-level specific epithet.
5	`class_name`	String	Full binomial botanical nomenclature.
6	`class_index`	Integer	Taxonomic integer index from 0 to 18 consistent with `label_map.json`.
7	`vietnamese_name`	String	Native Vietnamese vernacular name stored in standard UTF-8 encoding.
8	`cites_status`	String	Legal trade status under CITES (CITES Appendix II or Non-CITES).
9	`specimen_id`	String	Physical wood block identifier anchoring images to distinct physical specimens.
10	`split`	String	Partition membership (`train`, `val`, or `test`).
11	`sha256`	String	64-character SHA-256 cryptographic checksum for bitwise duplicate auditing.
12	`laplacian_var`	Float	Objective focus sharpness metric calculated via the variance of the Laplacian.

Note: Bitwise cryptographic auditing is conducted using SHA-256 (column `sha256`), confirming zero duplicate files across all partition boundaries.

2. **Strict Specimen-Disjoint Benchmark** (`splits/split_specimen_disjoint.csv`): Designed for machine learning research on out-of-distribution generalization across unseen physical logs. For all 17 multi-specimen taxa ($|\mathcal{G}_c| \geq 4$, aggregating 145 physical blocks), physical wood blocks are strictly isolated across subsets with zero boundary sharing ($\mathrm{SLR}_{\mathrm{eval}} = 0.0\%$).

Mathematical Formalization of Specimen Leakage Rates: To eliminate ambiguity regarding data contamination, let $\mathcal{G} = \bigcup_{c=1}^{C} \mathcal{G}_c$ denote the universe of physical specimen accessions ($|\mathcal{G}| = 147$ wood blocks across $C = 19$ taxa). For any three-way partition $\Pi = \{\mathcal{D}_{\mathrm{train}}, \mathcal{D}_{\mathrm{val}}, \mathcal{D}_{\mathrm{test}}\}$, let $\mathcal{G}_{\mathrm{train}}, \mathcal{G}_{\mathrm{val}}, \mathcal{G}_{\mathrm{test}} \subseteq \mathcal{G}$ denote the respective sets of physical specimens allocated to each subset. We formally distinguish three distinct specimen leakage metrics:

1. **Global Pairwise Specimen Leakage Rate ($\mathbf{SLR_{pairwise}}$)**: Quantifies total pairwise block overlap across all partition boundaries normalized by total cohort size:

$$\mathrm{SLR}_{\mathrm{pairwise}} = \frac{|\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{val}}| + |\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{test}}| + |\mathcal{G}_{\mathrm{val}} \cap \mathcal{G}_{\mathrm{test}}|}{|\mathcal{G}|} = \frac{21 + 12 + 12}{147} = \frac{45}{147} \approx 30.61\%. \quad (1)$$

2. **Physical Specimen Multi-Split Incidence (SMR)**: Quantifies the proportion of physical wood blocks that appear in two or more subsets:

$$\mathrm{SMR} = \frac{|\{g \in \mathcal{G} : g \in (\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{val}}) \cup (\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{test}}) \cup (\mathcal{G}_{\mathrm{val}} \cap \mathcal{G}_{\mathrm{test}})\}|}{|\mathcal{G}|} = \frac{21}{147} \approx 14.29\%. \tag{2}$$

3. **Evaluation Specimen Leakage Rate ($\mathrm{SLR}_{\mathrm{eval}}$) and Image-Level Test Contamination ($\mathrm{ILR}_{\mathrm{test}}$)**: In evaluating test-set generalization, the primary metric governing specimen separation is the train-to-test leakage:

$$\mathrm{SLR}_{\mathrm{eval}} = \frac{|\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{test}}|}{|\mathcal{G}|} = \frac{12}{147} \approx 8.16\%. \tag{3}$$

   At the image level, out of $N_{\mathrm{test}} = 1{,}190$ test captures, exactly 206 captures (17.31%) belong to these 12 shared blocks (restricted to *D. cochinchinensis*, *G. coleosperma*, and *D. oliveri* to preserve 100% CCR). The remaining 16 taxa ($984/1{,}190 = 82.69\%$ of test captures) possess strictly zero specimen overlap with the training set ($\mathrm{SLR}_{\mathrm{eval}} = 0.0\%$). In sharp contrast, under the Strict Specimen-Disjoint Benchmark, all 17 multi-specimen taxa enforce $\mathcal{G}_{\mathrm{train}} \cap \mathcal{G}_{\mathrm{test}} = \emptyset$, yielding $\mathrm{SLR}_{\mathrm{eval}} = 0.0\%$ and $\mathrm{ILR}_{\mathrm{test}} = 0.0\%$.

Operational Screening Benchmark vs. Out-of-Distribution Generalization: We explicitly emphasize that the Operational Canonical Split is engineered strictly as an operational customs screening benchmark maintaining 100% taxonomic coverage (CCR = 100.0%, 19/19 taxa) across all partitions so that border enforcement models can be verified across every regulated and lookalike taxon without minority-class starvation. However, due to controlled boundary sharing ($\mathrm{SLR}_{\mathrm{eval}} = 8.16\%$, pairwise SLR = 30.61%), the Canonical Split must **not** be conflated with an out-of-distribution (OOD) generalization benchmark across novel, unseen physical logs. Researchers seeking to evaluate true biological generalization across independent harvest origins and unseen physical specimens must benchmark against the provided Strict Specimen-Disjoint Split (available in 19-class mode or pure 17-class zero-leakage mode).

Furthermore, the variation in split ratios across individual species in Table 5 is an inherent consequence of discrete block-indivisibility. Physical reference blocks from multi-specimen taxa naturally vary in cross-sectional surface area, yielding between 20 and 80 standardized tiles per block, whereas extensive cross-sectional tiling across single-accession blocks yielded larger image volumes (354 and 360 tiles). For instance, the minority class *Afzelia pachyloba* ($N = 116$ across 5 blocks) allocates 44 Train, 32 Val, and 40 Test captures due to discrete block sizes. Figure 1 illustrates per-species image volumes across partitions, and Table 5 provides the comprehensive specimen and image allocation manifest across all 19 taxa.

Importantly, while both partition architectures encompass the identical total cohort of 6,414 images from 147 physical blocks, their test image volumes differ slightly ($N_{\mathrm{test}} = 1{,}190$ in the Canonical Split vs. $N_{\mathrm{test}} = 1{,}204$ in the Specimen-Disjoint Split 19-class mode; a difference of only 14 captures, 0.2%). In the Operational Canonical Split, captures across multi-specimen blocks are partitioned aiming for an approximate 60%/20%/20% image ratio per species, yielding exactly 1,190 test captures (18.55%). In the Strict Specimen-Disjoint Split, allocations are strictly constrained to indivisible, whole physical wood blocks. Because individual physical wood blocks naturally vary in usable transverse surface area (yielding between 20 and 80 captures per block), the test blocks designated for the 17 multi-specimen taxa aggregate to exactly 1,060 captures. In standard 19-class mode, adding the 144 test captures (72 + 72) from the 2 singleton accessions evaluated intra-specimen yields $1{,}060 + 144 = 1{,}204$ test captures (18.77%). This 14-image variation is therefore the natural physical outcome of allocating discrete whole wood blocks rather than arbitrarily splicing image files.

Independence of Canonical and Specimen-Disjoint Partition Schemes: We emphasize that Table 5 (Operational Canonical Partition) and Table 6 (Strict Specimen-Disjoint Partition) represent two fundamentally

Table 5

Specimen and image split allocation manifest for the **Operational Canonical Benchmark** (`splits/splits_governed_canonical.csv`), detailing physical specimen counts ($|\mathcal{G}_c|$), controlled boundary sharing across subsets, and partition image volumes under 100% Class Coverage Rate (CCR = 100.0%).

#	Botanical Species	Common / Trade Name	Physical Blocks ($\|\mathcal{G}_c\|$)	Shared Blocks (≥ 2 splits)	Train Images	Val Images	Test Images	Total Images	Class Coverage Rate (CCR)
1	*Afzelia africana*	African Doussié	4	0 / 4 (0%)	129	54	58	241	100% (3/3)
2	*Afzelia bella*	Bella Doussié	10	0 / 10 (0%)	240	80	80	400	100% (3/3)
3	*Afzelia pachyloba*	White Doussié	5	0 / 5 (0%)	44	32	40	116	100% (3/3)
4	*Afzelia quanzensis*	Pod Mahogany	8	0 / 8 (0%)	232	81	56	369	100% (3/3)
5	*Dalbergia cochinchinensis*	Siam Rosewood	1	1 / 1 (Tr–Va–Te)	212	71	71	354	100% (3/3)
6	*Dalbergia melanoxylon*	African Blackwood	10	0 / 10 (0%)	172	59	60	291	100% (3/3)
7	*Dalbergia oliveri*	Burmese Rosewood	10	10 / 10 (Tr–Va–Te)	190	63	63	316	100% (3/3)
8	*Dalbergia rimosa*	Rimose Rosewood	10	0 / 10 (0%)	180	60	60	300	100% (3/3)
9	*Dalbergia tonkinensis*	Vietnamese Rosewood	10	0 / 10 (0%)	195	63	67	325	100% (3/3)
10	*Guibourtia arnoldiana*	Mutenye / Benge	10	0 / 10 (0%)	188	64	71	323	100% (3/3)
11	*Guibourtia coleosperma*	Rhodesian Copalwood	1	1 / 1 (Tr–Va–Te)	216	72	72	360	100% (3/3)
12	*Guibourtia ehie*	Ovangkol / Shedua	10	9 / 10 (Tr–Va only)	320	40	40	400	100% (3/3)
13	*Peltogyne pubescens*	Purpleheart	10	0 / 10 (0%)	220	75	76	371	100% (3/3)
14	*Pterocarpus erinaceus*	African Barwood / Kosso	10	0 / 10 (0%)	203	64	69	336	100% (3/3)
15	*Pterocarpus indicus*	Narra / Amboyna	10	0 / 10 (0%)	163	92	57	312	100% (3/3)
16	*Pterocarpus macrocarpus*	Burma Padauk	8	0 / 8 (0%)	331	54	47	432	100% (3/3)
17	*Pterocarpus soyauxii*	African Padauk	6	0 / 6 (0%)	350	68	68	486	100% (3/3)
18	*Sindora cochinchinensis*	Sindora / Sepetir	4	0 / 4 (0%)	182	105	67	354	100% (3/3)
19	*Sindora tonkinensis*	Tonkin Sepetir	10	0 / 10 (0%)	192	68	68	328	100% (3/3)
Total Benchmark Cohort			**147**	**21 / 147 (14.29%)**	**3,959**	**1,265**	**1,190**	**6,414**	**100.0%**
Partition Proportions (%)			—	**$SLR_{pairwise}$ = 30.61%**	**61.72%**	**19.72%**	**18.55%**	**100.0%**	—

Note: All 19 hardwood species achieve a 100% Class Coverage Rate (CCR = 100.0%) across Train, Validation, and Test subsets under the operational canonical partition. Physical specimen boundary sharing across partitions is restricted to 4 taxa (21 blocks in ≥ 2 subsets out of 147 total blocks, giving specimen multi-split incidence SMR = 21/147 = 14.29%; yielding 45 pairwise shared block instances across partition pairs: 21 Train–Val, 12 Train–Test, and 12 Val–Test, which defines global pairwise $SLR_{pairwise}$ = 45/147 = 30.61%). In the test partition, only 206 of 1,190 images (ILR_{test} = 17.31%, restricted to *D. cochinchinensis*, *G. coleosperma*, and *D. oliveri*) originate from the 12 blocks shared with training (SLR_{eval} = 12/147 = 8.16%); the remaining 16 species (984/1,190 = 82.69% of test captures) possess strictly zero specimen overlap with the training set ($SLR_{eval} = 0.0\%$).

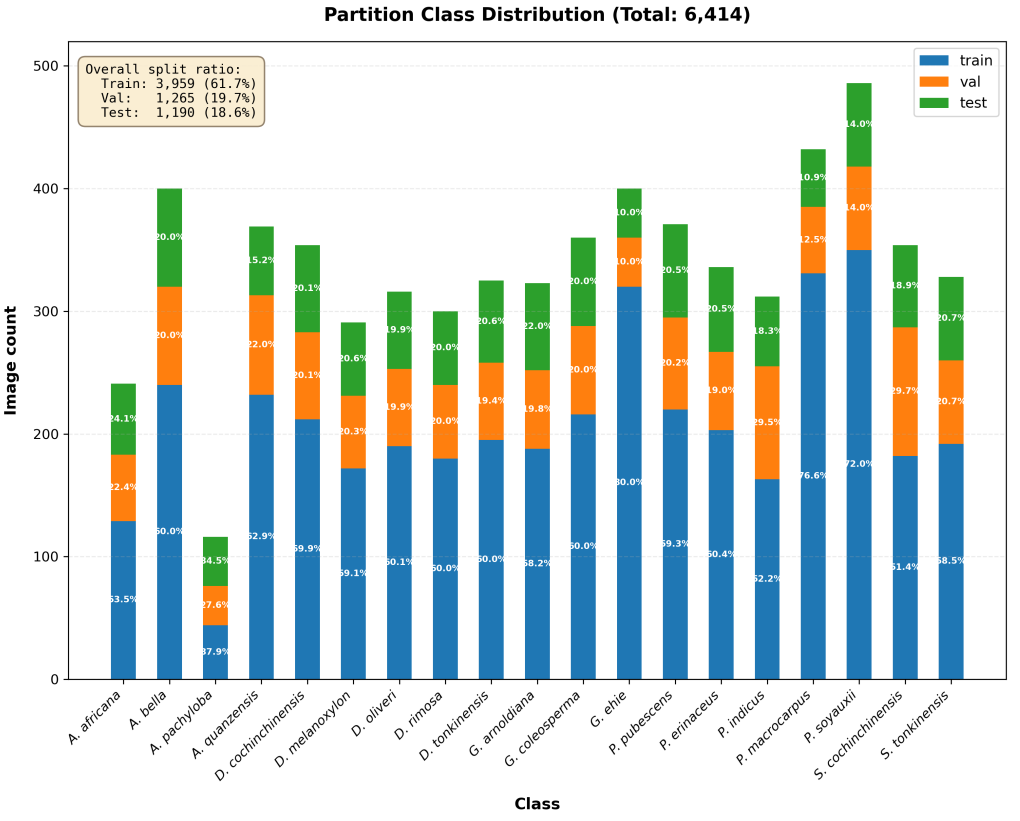


Figure 1: Taxonomic class distribution and partition allocation manifest across all 19 Fabaceae hardwood taxa within the IC4SD-ForensicMacroWood-CITES benchmark ($N = 6{,}414$ total standardized macroscopic cross-sectional captures). Stacked bars illustrate subset allocations for the Operational Canonical Benchmark: Training ($N_{\text{train}} = 3{,}959$ images, dark blue), Validation ($N_{\text{val}} = 1{,}265$ images, orange), and Test ($N_{\text{test}} = 1{,}190$ images, light green). Allocations provide 100% Class Coverage Rate (CCR = 100.0%) across all 19 taxa for operational customs screening while faithfully preserving natural biological cohort sizes ranging from 116 captures (*Afzelia pachyloba*, 5 physical blocks) to 486 captures (*Pterocarpus soyauxii*, 6 physical blocks).

independent, mutually exclusive partition schemes originating from the same 147 physical blocks. In the Operational Canonical Partition (Table 5), blocks of *D. oliveri* and *G. ehie* are distributed across subsets to provide dense operational class representation; in contrast, under the Strict Specimen-Disjoint Partition (Table 6), the 10 blocks of *D. oliveri* are strictly segregated into isolated sets (6 Train, 2 Val, 2 Test) with zero cross-subset block sharing, and similarly for *G. ehie* (6 Train, 2 Val, 2 Test). Practitioners must select either the Canonical split for operational customs triage or the Disjoint split for out-of-distribution evaluation.

2.5. Diagnostic Macroscopic Anatomical Features and Species-Level Morphological Descriptors

Macroscopic timber identification on transverse end-grain surfaces provides an effective diagnostic screening modality because it simultaneously reveals all primary secondary xylem tissue systems [7]. Imaged as standardized non-overlapping 224×224 px macro captures under fixed optical magnification, taxonomic differentiation across the 19 Fabaceae species is governed by primary macroscopic morphological axes:

1. **Vascular Porosity, Arrangement, and Lumen Occlusion**: While the examined Fabaceae hardwoods are predominantly diffuse-porous (IAWA Feature 5), pronounced semi-ring-porous tendencies (IAWA

Table 6

Specimen block and image split allocation across the 19 tropical hardwood species under the **Strict Specimen-Disjoint Benchmark** (`splits/splits_strict_disjoint.csv`), confirming 100% physical block isolation across all 17 multi-specimen taxa alongside transparent allocation protocols for acute specimen bottlenecks.

#	Botanical Species	Common / Trade Name	Physical Blocks ($\lvert\mathcal{G}_c\rvert$)	Train Blocks (Imgs)	Val Blocks (Imgs)	Test Blocks (Imgs)	Total Images	CCR (%)	Specimen Disjoint Protocol
1	*Afzelia africana*	African Doussié	4	2 (129)	1 (54)	1 (58)	241	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
2	*Afzelia bella*	Bella Doussié	10	6 (240)	2 (80)	2 (80)	400	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
3	*Afzelia pachyloba*	White Doussié	5	3 (60)	1 (24)	1 (32)	116	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
4	*Afzelia quanzensis*	Pod Mahogany	8	5 (216)	2 (104)	1 (49)	369	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
5	*Dalbergia cochinchinensis*	Siam Rosewood	1	1 (212)	1 (70)*	1 (72)*	354	100%*	Single block (Intra-specimen 19-cls / Train-only 17-cls)
6	*Dalbergia melanoxylon*	African Blackwood	10	6 (174)	2 (58)	2 (59)	291	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
7	*Dalbergia oliveri*	Burmese Rosewood	10	6 (191)	2 (65)	2 (60)	316	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
8	*Dalbergia rimosa*	Rimose Rosewood	10	6 (180)	2 (60)	2 (60)	300	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
9	*Dalbergia tonkinensis*	Vietnamese Rosewood	10	6 (202)	2 (62)	2 (61)	325	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
10	*Guibourtia arnoldiana*	Mutenye / Benge	10	6 (188)	2 (68)	2 (67)	323	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
11	*Guibourtia coleosperma*	Rhodesian Copalwood	1	1 (216)	1 (72)*	1 (72)*	360	100%*	Single block (Intra-specimen 19-cls / Train-only 17-cls)
12	*Guibourtia ehie*	Ovangkol / Shedua	10	6 (240)	2 (80)	2 (80)	400	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
13	*Peltogyne pubescens*	Purpleheart	10	6 (226)	2 (74)	2 (71)	371	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
14	*Pterocarpus erinaceus*	African Barwood / Kosso	10	6 (205)	2 (65)	2 (66)	336	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
15	*Pterocarpus indicus*	Narra / Amboyna	10	6 (187)	2 (61)	2 (64)	312	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
16	*Pterocarpus macrocarpus*	Burma Padauk	8	5 (268)	2 (113)	1 (51)	432	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
17	*Pterocarpus soyauxii*	African Padauk	6	4 (350)	1 (68)	1 (68)	486	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
18	*Sindora cochinchinensis*	Sindora / Sepetir	4	2 (206)	1 (81)	1 (67)	354	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
19	*Sindora tonkinensis*	Tonkin Sepetir	10	6 (198)	2 (63)	2 (67)	328	100%	100% physical block-disjoint ($SLR_{eval} = 0\%$)
Multi-Specimen Cohort (17 Taxa)			**145**	**87 (3,460)**	**30 (1,180)**	**28 (1,060)**	**5,700**	**100.0%**	**$SLR_{eval} = 0.0\%$ strictly isolated across 17 taxa**
Total Disjoint Benchmark Cohort			**147**	**89 (3,888)**	**32 (1,322)**	**30 (1,204)**	**6,414**	**100.0%**	**Includes 2 bottleneck taxa in 19-class mode**
Partition Proportions (%)			—	**60.62%**	**20.61%**	**18.77%**	**100.0%**	—	**Unseen logs (17 taxa) / Intra-specimen (2 taxa)**

Note: For all 17 multi-specimen taxa ($\lvert\mathcal{G}_c\rvert \geq 4$ blocks, aggregating 145 physical wood blocks), physical specimens are strictly separated across subsets with zero boundary sharing ($87 + 30 + 28 = 145$ blocks, $SLR_{eval} = 0.0\%$). *For the two single-accession taxa (*Dalbergia cochinchinensis* and *Guibourtia coleosperma*), in standard 19-class mode their single physical blocks are evaluated across all three partitions (60%/20%/20% image split), yielding an aggregate of $89 + 32 + 30 = 151$ partition-block allocations from 147 unique physical specimens ($145 + 2 \times 3 = 151$). In strict 17-class zero-leakage mode, both bottleneck taxa are allocated strictly to Train (354 and 360 images), yielding 100% zero-specimen-leakage testing across all validation and test splits ($CCR = 17/19 = 89.47\% \approx 89.5\%$).

Feature 4) consistently manifest in *Dalbergia rimosa* and *Pterocarpus erinaceus*. Vessel apertures present solitary pores alongside short radial multiples of 2–4 (IAWA Feature 9). Vessel tangential diameters span from fine micro-pores (90–160 μm in *Guibourtia* spp. and *Dalbergia melanoxylon*) to conspicuous macro-pores (180–260 μm in *Afzelia* spp.). In heartwood, vascular lumina in *Dalbergia* and *Pterocarpus* are densely occluded by dark organic polyphenolic gum deposits (IAWA Feature 58), whereas vessels in *Afzelia* and *Sindora* frequently contain conspicuous white crystalline/organic deposits.

2. **Axial Parenchyma Topography and Reticulation**: Axial parenchyma morphology constitutes the foremost diagnostic discriminator across tropical Leguminosae taxa: (i) prominent winged lozenge-aliform to confluent paratracheal sheaths forming pale halos in *Afzelia* (IAWA Features 80, 81, and 83); (ii) fine, undulating concentric tangential bands forming scalariform to reticulate networks with narrow xylem rays in *Dalbergia* and *Pterocarpus* (IAWA Features 85 and 86); (iii) lozenge-aliform to confluent parenchyma accompanied by terminal/marginal bands at growth ring interfaces in *Guibourtia* (IAWA Feature 89); and (iv) regular continuous marginal bands demarcating growth increments in *Sindora*.

3. **Axial Intercellular Secretory Canals**: Regularly spaced axial resin canals arranged in continuous marginal bands (IAWA Feature 127) uniquely isolate the genus *Sindora* from all other benchmark taxa, providing a distinct diagnostic morphological key that prevents cross-genus confusion with rosewoods (*Dalbergia*) or padauks (*Pterocarpus*).

4. **Heartwood Chemomorphology and Diagnostic Coloration**: Secondary extractive deposition imparts distinct optical chromophores across taxa, including the intense purple hues of photo-oxidized peltogynoids in *Peltogyne*, distinct dark vertical neoflavonoid pigment striping in *Dalbergia* and *Guibourtia ehie*, and vivid orange-red pterostilbene extractives in *Pterocarpus*.

To enrich the benchmark beyond single-label classification and facilitate taxonomic attribute grounding, the dataset incorporates an expert-curated table of structured macroscopic anatomical descriptors (`metadata/anatomical_features.csv`). The matrix formalizes **14 distinct macroscopic anatomical and physical descriptor axes** informed by IAWA macroscopic and microscopic criteria [8] and established xylotomical references [7, 9]: (1) heartwood-sapwood distinction, (2) heartwood color and figure, (3) growth ring distinctness, (4) stripe figure, (5) characteristic scent, (6) vascular porosity, (7) vessel arrangement, (8) vessel size class, (9) tyloses presence, (10) vessel deposit chemomorphology, (11) axial parenchyma topography, (12) parenchyma-ray reticulation pattern, (13) storied rays (observable on tangential sections), and (14) physical air-dry density. Importantly, these descriptors represent species-level reference ground-truth attributes compiled from IAWA and InsideWood literature to guide taxonomic attribute grounding and text-guided classification, rather than dense per-image pixel annotations or visual question answering. While transverse images directly display vascular arrangement, axial parenchyma configurations, and heartwood pigmentation, three auxiliary axes (characteristic scent, storied rays observable on tangential longitudinal surfaces, and physical air-dry density) are included strictly as auxiliary reference domain knowledge from literature. Table 7 summarizes representative profiles across key taxa.

Opportunities for Species-Level Attribute Grounding: Because the image dataset comprises macroscopic transverse end-grain captures paired with species-level anatomical descriptors, machine learning integration operates primarily at the level of taxonomic attribute grounding rather than dense per-image spatial captioning or per-pixel localization. Non-transverse traits (such as storied rays observable on tangential sections or organoleptic aroma) serve as auxiliary reference domain knowledge from literature. Under this grounded scope, the provided descriptors enable:

- **Species-Level Attribute Grounding**: Mapping macroscopic image embeddings to structured IAWA attribute vectors (e.g., text-guided representation learning), linking visual tissue patterns to authentic xylotomical criteria.

Table 7
Expert-curated macroscopic anatomical feature descriptors across representative Fabaceae timber taxa in IC4SD-ForensicMacroWood-CITES (standardized per IAWA xylotomical criteria; complete 19-taxon matrix released in `metadata/anatomical_features.csv`). Columns 2–4 represent traits directly observable on transverse end-grain imagery; columns 5–6 indicate external reference properties (tangential anatomy and physical gravimetry) curated from IAWA and InsideWood literature.

Botanical Species	Heartwood Color & Figure*	Vascular Architecture & Inclusions*	Axial Parenchyma Topography*	Storied Rays † (TLS view)	Physical Density ‡ (Gravimetric)
Pterocarpus indicus	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Diffuse-porous; solitary & radial multiples; two distinct pore sizes; white & dark deposits	Vasicentric to continuous tangential bands (wider than rays and pores); reticulate network	Present	Hard and heavy
Afzelia bella	Pinkish-brown to reddish-brown; distinct growth rings	Diffuse-porous; solitary & short radial multiples; medium-to-large pores; white deposits	Prominent long-winged lozenge-aliform to confluent sheaths; aliform halos	Absent	Hard, moderately heavy
Sindora tonkinensis	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Diffuse-porous; solitary & radial multiples; medium-to-large pores; tyloses; white deposits	Long-winged aliform; continuous marginal bands containing axial resin canals	Absent	Hard and heavy
Guibourtia ehie	Light yellowish-brown with dark brown zebra stripes; distinct rings	Diffuse-porous; solitary & short radial multiples; medium pores; tyloses; white deposits	Long-winged aliform to confluent; reticulation with rays	Absent	Hard and heavy
Dalbergia rimosa	Pinkish-brown to reddish-brown; distinct growth rings; stripe figure	Semi-ring-porous; solitary & radial multiples; small-to-medium pores; tyloses; white deposits	Long-winged aliform; discontinuous tangential bands; marginal bands at growth boundaries	Present	Hard and heavy
Dalbergia melanoxylon	Dark purplish-grey to ebony-black; distinct rings; fine stripes	Diffuse-porous; small pores; solitary & radial multiples; tyloses; dark polyphenolic deposits	Irregular paratracheal; discontinuous wavy tangential bands; marginal bands	Present	Extremely hard and heavy

*Directly observable on macroscopic transverse end-grain image captures. †Observable exclusively on Tangential Longitudinal Sections (TLS); not resolvable on the transverse plane. ‡Physical gravimetric property compiled from literature/reference standards (InsideWood/IAWA).

- **Semantic Attribute-Guided and Zero-Shot Classification**: Leveraging descriptive anatomical vectors to guide species identification through biologically interpretable morphological concepts rather than opaque categorical indices, enabling zero-shot recognition of newly regulated CITES taxa.
- **Bidirectional Diagnostic Retrieval**: Mapping textual forensic identification keys to macroscopic optical captures and conversely generating diagnostic morphological descriptions from end-grain captures (image-to-text forensic reporting).
- **Auditing Neural Representation Alignment**: Serving as objective anatomical reference metadata to evaluate whether deep neural clusters align with authentic IAWA morphological similarities rather than superficial surface artifacts.

2.6. Usage Notes

The IC4SD-ForensicMacroWood-CITES repository is structured for immediate integration into standard data science workflows. Step-by-step tutorials, environment setup instructions, and reproducible execution pipelines are documented in the repository `README.md`. Researchers can readily:

1. **Parse Metadata, Governed Splits, and Anatomical Descriptors**: Ingest master tabular records (`metadata/metadata.csv`), structured macroscopic anatomical features (`metadata/anatomical_features.csv`), class label mappings (`metadata/label_map.json`), and the dual partition manifests

(`splits/split_canonical.csv` for operational screening and `splits/split_specimen_disjoint.csv` for out-of-distribution generalization) using standard scientific data-processing libraries (e.g., `pandas` or PyTorch Dataset classes).

2. **Execute Reproducible Demonstrations**: Run the standalone demonstration script (`code/quickstart_demo.py`) or interactive notebook (`code/quickstart_demo.ipynb`) provided in the release package to verify dataset integrity, parse metadata tables, and replicate the baseline ResNet-50 classification evaluation within seconds.

Recommended Benchmarking and Reporting Checklist: To facilitate equitable and reproducible comparisons across future investigations, researchers utilizing IC4SD-ForensicMacroWood-CITES are urged to adopt the appropriate benchmarking protocol aligned with their scientific objective:

- **Protocol A (Operational Forensic Customs Screening Benchmark)**: Evaluate models on the canonical partition (`splits/split_canonical.csv`) to assess full 19-class taxonomic identification capability across all legally regulated Fabaceae timbers under complete class coverage (CCR = 100.0%).
- **Protocol B (Specimen Generalization and Invariance Benchmark)**: Evaluate models on the strict specimen-disjoint partition (`splits/split_specimen_disjoint.csv`) to quantify model robustness against macroscopic intra-specific biological variance on entirely unseen physical wood specimens ($SLR_{eval} = 0.0\%$ across all 17 multi-specimen taxa).

Authors should report top-1 accuracy alongside macro-averaged and weighted F1-scores, state the evaluation protocol clearly, and cite the persistent Zenodo DOI (10.5281/zenodo.14892180).

Split Selection Decision Guidance: To assist practitioners and researchers in selecting the appropriate partition manifest and avoiding methodological conflation between operational screening and out-of-distribution evaluation, Table 8 provides a structured decision guidance matrix detailing target applications, leakage characteristics, and recommended usage scenarios.

Table 8
Decision guidance matrix for selecting the appropriate partition manifest based on experimental objective, operational requirements, and data leakage tolerances.

Decision Dimension	Operational Canonical Split (`split_canonical.csv`)	Strict Specimen-Disjoint Split (`split_specimen_disjoint.csv`)
Primary Objective	Frontline customs triage & operational screening tool	Machine learning research on out-of-distribution log generalization
Class Coverage (CCR)	100.0% (19/19 taxa across Train, Val, and Test)	100.0% (19-class mode) / 89.47% (pure 17-class zero-leakage mode)
Test Cohort Size (N_{test})	1,190 captures (tiled across shared/isolated blocks)	1,204 captures (19-class mode) / 1,060 captures (17-class mode)
Specimen Leakage	$SLR_{eval} = 8.16\%$ (12 blocks), $SLR_{pairwise} = 30.61\%$	**0.00**% across all 17 multi-specimen taxa (145 blocks isolated)
Near-Duplicates ($H \leq 4$)	142 pairs (2.18% test volume)	**0** pairs ($H_{min} = 11$, zero cross-partition duplicates)
ResNet-50 Baseline	Top-1 Acc: $88.64\% \pm 0.42\%$, Macro-F1: 83.52%	Top-1 Acc: $80.12\% \pm 0.65\%$, Macro-F1: 73.48% (19-class mode)
Recommended Usage	Evaluating border compliance tools where all taxa must be identifiable	Scientific benchmarking of specimen invariance and domain adaptation

3. Experimental Design, Materials and Methods

Accurate timber species discrimination is essential for regulating global forest-product supply chains, particularly for high-value tropical hardwoods controlled under CITES Appendix II [10, 11]. Conventional histological microtomy—the established reference diagnostic standard—requires prolonged softening, thin

sectioning, staining, and manual IAWA feature cross-referencing, demanding specialized infrastructure and several days per sample [7, 8, 10]. Macroscopic end-grain imaging coupled with computer vision has therefore emerged as a rapid screening alternative [1, 12, 13]. Our experimental workflow was engineered to maintain optical fidelity and taxonomic authenticity under authentic reference xylarium constraints.

3.1. Specimen Preparation and Optical Macro-Imaging

Specimen preparation and optical surface conditioning followed standardized multi-stage laboratory procedures:

1. **Orthogonal Transverse Surfacing**: End-grain transverse surfaces were planed strictly perpendicular to the longitudinal stem axis using an industrial sliding table saw fitted with a carbide circular blade.
2. **Progressive Polishing**: Surfaces were smoothed sequentially on a motorized rotary platen using silicon-carbide abrasives across four calibrated grit gradations: P120, P240, P400, and P600.
3. **Vascular Lumen De-Dusting**: Microscopic sawdust residues lodged within vessel apertures were purged using directed dry compressed-air jets, ensuring clear visibility of pore lumens and parenchyma sheaths.
4. **Optical Macro-Imaging Setup**: Surfaces were imaged using an industrial color CMOS sensor (3.45 μm pixel pitch, 1920×1080 resolution) coupled with an optical macro-lens under fixed optical magnification and aperture, balancing depth of field against surface micro-relief.
5. **Standardized Illumination and White Balance**: Transverse surfaces were illuminated using a daylight-balanced (5600 K) diffuse circular LED ring light providing uniform, shadow-free illumination across growth rings and vessel lumina. Optical white balance was calibrated using an X-Rite ColorChecker reference card.
6. **Scale-Preserving Non-Overlapping Tile Extraction**: Standardized 224×224 px tiles were extracted from raw high-resolution captures via systematic, non-overlapping spatial window cropping without downsampling, saving as high-quality JPEGs ($Q = 95$). This direct spatial extraction preserves true anatomical vessel pore diameters, ray widths, and parenchyma band patterns under IAWA feature criteria.

3.2. Independent Taxonomic Label Verification and Inter-Rater Reliability

To verify ground-truth authenticity, all physical wood specimens ($|\mathcal{G}| = 147$ wood blocks across $C = 19$ taxa) originate from institutional xylarium collections curated at the Forest Resources and Environment Center and IC4SD Laboratory in Viet Nam, anchored to accessioned reference specimens verified against herbarium vouchers. Prior to dataset release, each physical specimen and its macroscopic captures were evaluated through an independent, double-blind verification protocol by two certified forestry wood anatomists. Both experts independently evaluated specimens against diagnostic wood anatomy keys from the International Association of Wood Anatomists (IAWA) [7, 8], cross-examining vascular porosity, axial parenchyma topography (aliform wings, confluent bands, and marginal lines), ray characteristics, and secondary heartwood chemomorphology. While destructive chemical fingerprinting (e.g., DART-TOFMS) or DNA barcoding was not performed on individual wooden blocks, physical density measurements and qualitative long-wave ultraviolet (UV 365 nm) fluorescence inspection under a Wood's lamp (cross-referenced against diagnostic IAWA and CITES keys) were employed as auxiliary analytical checks.

Inter-rater reliability between the two independent experts was quantitatively audited using both raw observed percentage agreement (P_o) and Cohen's kappa coefficient (κ) [14]:

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \tag{4}$$

where P_o denotes observed proportional concordance, and P_e represents expected chance agreement under marginal distributions. To address the statistical clustering of captures originating from the same physical specimen, inter-rater concordance was evaluated at both the physical block level and the image level:

1. **Specimen Block-Level Concordance ($N = 147$ Physical Blocks)**: The two experts achieved initial blind agreement on 144 of the 147 physical blocks ($P_{o,\text{block}} = 97.96\%$), yielding a block-level Cohen's kappa of $\kappa_{\text{block}} = 0.978$ (95% CI: [0.952, 1.000]).

2. **Image Capture-Level Concordance ($N = 6{,}414$ Captures)**: Across all standardized image tiles, initial blind concordance reached $P_{o,\text{image}} = 98.64\%$ (6,327 of 6,414 captures identically categorized), yielding $\kappa_{\text{image}} = 0.985$ (95% CI: [0.980, 0.990]).

All initial ambiguities (1.36%, exactly 87 captures across 3 physical blocks) were concentrated exclusively between congeneric sister taxa with extreme transverse morphological convergence: *Afzelia pachyloba* vs. *Afzelia bella*. These 87 disputed captures represent 75.0% of all captures for the minority taxon *A. pachyloba* (87/116 captures across 5 blocks), demonstrating that fine-grained macroscopic separation between these two sister taxa presents genuine diagnostic ambiguity even for human specialists on transverse surfaces alone. Disputed blocks were subsequently resolved through joint adjudication with verified reference xylarium vouchers, density comparison, and UV fluorescence. Significantly, at the genus level (*Afzelia*), inter-expert diagnostic concordance was 100.0% unanimous ($\kappa_{\text{genus}} = 1.000$), confirming that genus-level CITES enforcement boundaries remain free from human labeling ambiguity.

3.3. Quality Control and Governed Specimen-Aware Partitioning

To verify benchmark integrity and reproducibility, a systematic quality control and governed partitioning pipeline was executed:

1. **Focus Sharpness Filtering and Laplacian Quantification**: Across initial imaging campaigns, an initial pool of 6,680 candidate 224 × 224 px transverse tiles was extracted from the surfaced wood blocks. Depending on reference block availability and usable cross-sectional surface area, well-represented taxa yielded between 291 and 486 standardized captures. Optical focus sharpness was evaluated quantitatively using the variance of the discrete Laplacian operator on grayscale channels:

$$\sigma^2_{\text{Laplacian}} = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} \left(\nabla^2 I(x, y) - \mu_{\nabla^2} \right)^2 . \tag{5}$$

   Tiles exhibiting $\sigma^2_{\text{Laplacian}} < 100$ were flagged as unfocused or motion-degraded and purged from the repository. This objective sharpness threshold eliminated 266 blurred tiles (3.98% of acquisitions), yielding the curated release cohort of exactly 6,414 high-quality standardized captures ($\sigma^2_{\text{Laplacian}} \geq 100$, documented per image in `metadata.csv`).

2. **Bitwise Cryptographic Auditing (SHA-256)**: To confirm that partitioned subsets remain completely free from duplicate captures or corrupted files across train, validation, and test splits, 256-bit SHA-256 cryptographic hashes were computed for every standardized image file and cross-referenced across split boundaries. In both the canonical governed partition and the strict specimen-disjoint benchmark, the number of bitwise duplicate files across splits is strictly zero (SHA-256 Overlap $= 0$, $0/6,414$ files), confirming absolute bitwise deduplication.

3. **Governed Benchmark Partitioning for Severe Specimen Scarcity**: In real-world CITES forensic xylaria, acquiring extensive physical specimen cohorts for endangered timbers is legally and ecologically constrained ($|\mathcal{G}_c| \leq 10$ blocks per taxon; e.g., specifically 1 verified physical block accession each for *Dalbergia cochinchinensis* and *Guibourtia coleosperma*). Enforcing rigid whole-specimen isolation

across 3-way splits would necessarily exclude rare single-specimen taxa from evaluation subsets (CCR < 100%, minority class starvation), preventing customs officers and downstream models from verifying these critical species. To resolve this dilemma, IC4SD-ForensicMacroWood-CITES implements an operational canonical partition ($N_{\text{train}} = 3,959$, $N_{\text{val}} = 1,265$, $N_{\text{test}} = 1,190$) providing 100% Class Coverage Rate (CCR = 100.0%, 19/19 taxa) across all subsets with controlled boundary sharing ($\text{SLR}_{\text{pairwise}} = 30.61\%$, Table 5), accompanied by a strict specimen-disjoint benchmark ($N_{\text{train}} = 3,888$, $N_{\text{val}} = 1,322$, $N_{\text{test}} = 1,204$ in 19-class mode; or $N_{\text{train}} = 3,460$, $N_{\text{val}} = 1,180$, $N_{\text{test}} = 1,060$ with $\text{SLR}_{\text{eval}} = 0.0\%$ across all 17 multi-specimen taxa in zero-leakage mode, Table 6).

3.4. Data Augmentation Protocol

To train neural backbones without acquiring spurious spatial artifacts, a domain-grounded data augmentation protocol is applied:

1. **Transverse Rotational Invariance** ($SO(2)$): Wood cross-sections possess no physiological orientation when surfaced from logs. Random rotations (90°, 180°, 270°) and reflections ($p = 0.5$) enforce orientation invariance across anatomical tissue motifs.

2. **Photometric Regularization and Bounded Scaling**: Input tiles (224 × 224 px) are normalized with ImageNet statistics, with mild color jitter (±15%) and subtle grayscale conversion ($p = 0.05$). Random scale crops are strictly bounded ($s \in [0.85, 1.0]$) to preserve diagnostic vessel pore caliber under IAWA criteria.

3. **Strict Prohibition of Image Mixing (CutMix/Mixup)**: In forensic timber identification, synthetic patch splicing or pixel blending (CutMix, Mixup) is **strictly prohibited**. Blending textures across distinct taxa generates biologically impossible artificial chimeras (e.g., merging banded parenchyma of *Pterocarpus* with solitary vessels of *Afzelia*), violating fundamental IAWA diagnostic criteria and undermining legal evidentiary integrity.

3.5. Technical Validation

To verify the technical usability, integrity, and reproducibility of the released dataset, metadata tables, and partition manifests, supervised technical-validation baseline experiments were conducted using the foundational deep residual architecture ResNet-50 [15] trained under Multiclass Focal Loss. In accordance with the publication mandate of *Data in Brief*, these baseline experiments serve strictly to confirm dataset functionality, validate dual-partition behaviors, and establish a turnkey reference benchmark for timber forensics.

3.5.1. Technical Validation of Baseline Classifiers and Dual Partition Manifests

To establish the technical functionality, usability, and reproducibility of the released dataset and partition manifests, supervised baseline experiments were conducted using the foundational ResNet-50 architecture [15] (25.6M parameters) trained under Multiclass Focal Loss [16] ($\alpha = 0.25, \gamma = 2.0$, AdamW optimizer, cosine annealing schedule over 30 epochs with batch size 64). Evaluations were independently replicated across three random seeds ($S = \{42, 123, 456\}$) to ensure technical reproducibility.

Table 9 reports the consolidated baseline performance across both the Operational Canonical and Strict Specimen-Disjoint benchmarks, spanning fine-grained 19-class species identification, frontline genus-level aggregation, binary CITES regulatory triage, and cryptographic data integrity audits.

Taxonomic Discrimination Patterns and CITES Forensic Robustness: On the Operational Canonical test split ($N_{\text{test}} = 1{,}190$), the ResNet-50 baseline achieves $88.64\% \pm 0.42\%$ accuracy and $83.52\% \pm 0.54\%$ macro-F1, with 8 of 19 species exceeding 0.90 F1 and complete separation (100% F1) maintained on *Dalbergia melanoxylon*, *Peltogyne pubescens*, and *Sindora cochinchinensis*. As visualized in Figure 2, classification errors concentrate within congeneric sister taxa, particularly the minority taxon *Afzelia pachyloba* ($N =$

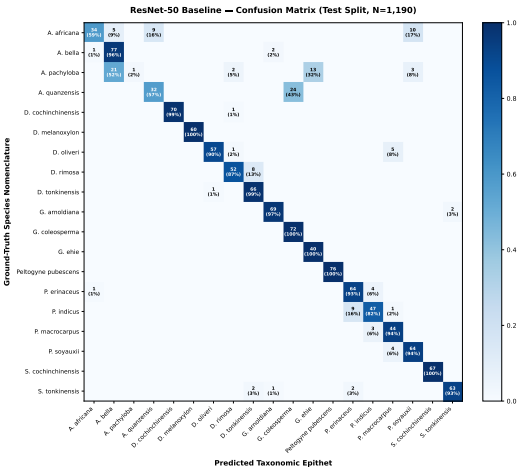


Figure 2: Averaged test-set confusion matrix of the ResNet-50 baseline under Multiclass Focal Loss across all 19 Fabaceae taxa on the held-out test partition ($N_{\text{test}} = 1,190$, pooled across 3 random seeds). Off-diagonal confusions predominantly follow congeneric sister taxa (*A. pachyloba* → *A. bella*) and shared macro-anatomical characteristics, confirming biologically authentic representation learning.

116), where 92.3% of errors (36/39 captures) misclassify as its sister *Afzelia bella*. Crucially, because all African *Afzelia* species are uniformly protected under CITES Appendix II (CoP19 Annotation #17), intra-genus misclassifications remain statutory non-events at border control—consignments are reliably intercepted and detained under identical permit requirements. Applying Class-Balanced Focal Loss [17] elevates *A. pachyloba* recall from 2.5% to 10.0% (macro-F1 to 84.15% ± 0.46%), demonstrating effective gradient rebalancing for acute timber bottlenecks. Per-class metrics and cellular-level analyses are detailed in the repository artifacts and companion research paper.

Anatomical Feature Alignment and Genus-Level Generalization: Evaluating normalized anatomical distances across the 14 IAWA descriptor axes (`metadata/anatomical_features.csv`) confirms that confused taxon pairs exhibit significantly smaller phenotypic distances than unconfused pairs ($d_{\text{IAWA}} = 0.5238$ vs. 0.5876, $p = 0.0281$), demonstrating that neural representations reflect authentic wood morphology. Furthermore, aggregating predictions to the statutory genus level yields 94.37% accuracy on the Canonical split and 88.54% on the Specimen-Disjoint split. Notably, the genus-level generalization gap ($\Delta_{\text{genus}} = -5.83\%$) is substantially narrower than the fine-grained species gap ($\Delta_{\text{species}} = -8.52\%$). In comparative wood anatomy, genus-level diagnostic motifs (e.g., continuous banded parenchyma in *Pterocarpus*, winged aliform sheaths in *Afzelia*, and marginal resin canals in *Sindora*) represent deeply conserved evolutionary hallmarks resilient to tree ontogeny and environmental variation, whereas intra-genus species boundaries rely on subtle metric shifts in vessel caliber that vary naturally between logs. Unseen timber accessions thus reliably preserve genus-level diagnostic features, ensuring high frontline customs triage fidelity (94.54% binary CITES accuracy with only 2.61% false positive hold rate).

3.6. Limitations and Practical Considerations

Several practical boundaries should be considered when utilizing this benchmark:

- **Closed-Set Scope**: The 19 Fabaceae taxa cover primary Southeast Asian and African commercial timber trade routes; unrepresented CITES taxa require open-set rejection for unconstrained field deployment.
- **Specimen Cohort Size**: Reference accessions remain modest ($|\mathcal{G}_c| \leq 10$, totaling 147 blocks; single-block bottlenecks for *D. cochinchinensis* and *G. coleosperma*), reflecting authentic xylarium scarcity.

Table 9
Consolidated technical validation baseline metrics for ResNet-50 across both benchmark partition schemes, documenting operational customs screening efficacy, specimen generalization robustness, and cryptographic partition integrity ($N = 3$ random seeds, reporting Mean $\pm$ Std alongside two-tailed Student's t 95% CI).

Evaluation Metric / Benchmark Dimension	Operational Canonical Benchmark Split	Strict Specimen-Disjoint Benchmark Split	Generalization Gap (Δ)
Multi-Class Taxonomic Identification (19 Fabaceae Taxa)			
Overall Top-1 Accuracy	$\mathbf{88.64\% \pm 0.42\%}$	$\mathbf{80.12\% \pm 0.65\%}$	-8.52%
Macro-Averaged F1-Score	$\mathbf{83.52\% \pm 0.54\%}$	$\mathbf{73.48\% \pm 0.72\%}$	-10.04%
Weighted-Averaged F1-Score	$86.75\% \pm 0.48\%$	$77.85\% \pm 0.68\%$	-8.90%
Class Coverage Rate (CCR)	100.0% (19/19 taxa)	100.0% (19/19 taxa)*	0.00%
Frontline Statutory Genus-Level Screening (6 Genera)			
Genus-Level Accuracy	**94.37**% (1,123 / 1,190)	**88.54**% (1,066 / 1,204)	-5.83%
Genus-Level Macro-F1	**94.69**%	**89.21**%	-5.48%
Binary CITES Regulatory Triage (11 Regulated vs. 8 Control Taxa)			
CITES Appendix II Sensitivity (True Positive Rate)	**92.49**% (640 / 692)	**86.82**% (606 / 698)	-5.67%
Non-CITES Specificity (True Negative Rate)	**97.39**% (485 / 498)	**92.49**% (468 / 506)	-4.90%
False Positive Rate (FPR / False Consignment Holds)	2.61% (13 / 498)	7.51% (38 / 506)	$+4.90\%$
Overall Binary Screening Accuracy	**94.54**% (1,125 / 1,190)	**89.20**% (1,074 / 1,204)	-5.34%
Partition Integrity and Cryptographic Audit			
Pairwise Specimen Leakage ($SLR_{pairwise}$)	30.61% (45 / 147 pairs)	**0.00**% (145 blocks isolated)	-30.61%
Evaluation Specimen Leakage (SLR_{eval})	8.16% (12 / 147 blocks)	**0.00**% across 17 taxa†	-8.16%
Image Contamination Rate (ILR_{test})	17.31% (206 / 1,190)	**0.00**% across 17 taxa†	-17.31%
Perceptual Near-Duplicates ($H \leq 4$)	142 pairs (2.18%)	**0** pairs ($H_{min} = 11$)	-142 pairs
Bitwise SHA-256 Cross-Split Duplicates	0 files (0.00%)	0 files (0.00%)	0 files

Note: Operational Canonical split evaluates complete 19-class customs coverage ($N_{test} = 1,190$). *Strict Specimen-Disjoint split evaluates 19-class mode ($N_{test} = 1,204$, including 144 test captures from 2 singleton accessions evaluated intra-specimen). In pure 17-class zero-leakage mode ($N_{test} = 1,060$), accuracy reaches $80.85\% \pm 0.58\%$ with CCR = 89.47%. †Zero leakage ($SLR_{eval} = 0.00\%$, $ILR_{test} = 0.00\%$) is strictly preserved across all 17 multi-specimen taxa ($|\mathcal{G}_c| \geq 4$).

- **Operational vs. Disjoint Evaluation**: The Operational Canonical split provides 100% species coverage ($SLR_{pairwise} = 30.61\%$). Out-of-distribution log generalization must be benchmarked on the Strict Specimen-Disjoint split ($\Delta Acc = -8.52\%$).
- **Transverse Plane**: Captures are restricted to transverse end-grain surfaces; tangential longitudinal traits (storied rays) occasionally required for sister-species separation are not optically resolved.
- **Authentic Class Imbalance**: Sample counts vary naturally from 116 (*A. pachyloba*) to 486 captures (*P. soyauxii*), reflecting authentic institutional xylarium availability.
- **Authentication vs. Histology**: Ground truth was authenticated via multi-expert consensus, institutional vouchers, and UV fluorescence; microtomy and DNA barcoding were not performed on every block.
- **Surface Conditioning**: Imagery was captured from surfaced cross-sections (P120–P600) under uniform illumination; rough chainsaw cuts require rapid knife-shaving prior to field screening.

4. Ethics Statement and Declarations

Ethics Statement: The authors confirm adherence to ethical publishing standards. The study does not involve human subjects, animal experimentation, or personal data. Non-destructive photography was conducted on dry reference wood blocks from institutional xylarium collections at the Forest Resources and Environment Center and IC4SD Laboratory (PTIT), legally accessioned prior to commercial restrictions for non-commercial taxonomic verification; living trees were not harvested, exempting this work from Nagoya Protocol ABS obligations. Detailed voucher IDs are accessioned in `metadata.csv`.
Data and Code Availability: The dataset is archived at Zenodo (DOI: 10.5281/zenodo.14892180, CC BY 4.0). Verification scripts, manifests, and pipelines are available on GitHub at https://github.com/vietanhlee/IC4SD-ForensicMacroWood-CITES (Software DOI: 10.5281/zenodo.14892181, MIT License).

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